Pros

- Provides granular control over quantum circuit operations.
- Strong support for noise modelling and simulation.
- Supported by Google's extensive research infrastructure.

Cons

- Primarily focused on Google's quantum ecosystem.
- May require a steep learning curve for those new to quantum computing.

PyQuil

This open source quantum programming library has been developed by Rigetti Computing. It is part of the Forest SDK and is designed to facilitate the development of quantum-classical hybrid algorithms. PyQuil provides tools for writing and simulating quantum programs using the Quil language, and it supports execution on Rigetti's quantum hardware. It is particularly suited for researchers interested in exploring practical quantum applications.

Website: *https://github.com/rigetti/pyquil*

Technical features

- *Quil compiler:* Translates high-level quantum instructions into executable code.
- *Quantum-classical hybrid algorithms:* Supports the development of algorithms that leverage both quantum and classical resources.
- *Rigetti's Quantum SDK:* Provides access to Rigetti's quantum hardware and simulators.

Pros

- Strong focus on hybrid quantum-classical workflows.
- Provides access to Rigetti's quantum devices and simulators.
- Comprehensive documentation and active community support.

Cons

- Limited to Rigetti's quantum ecosystem.
- May require familiarity with the Quil programming language.

Hybrid quantum machine learning

Hybrid QML combines quantum computing with classical machine learning to leverage the strengths of both paradigms. The aim is to enhance computational efficiency and improve the performance of machine learning models by using quantum resources wherever they offer an advantage.

Quantum feature mapping: Quantum computers are used to transform classical data into a higher-dimensional quantum space, where it may become easier to classify or analyse it using classical algorithms.

Quantum circuits encode classical data into quantum states, while classical algorithms process the resulting quantum-enhanced data.

Quantum kernel estimation: Quantum computers calculate kernel functions, which measure the similarity between data points in a quantum-enhanced feature space, used in classical machine learning models like support vector machines.

Here, quantum circuits compute complex kernel matrices while classical algorithms utilise these matrices for learning tasks.

Variational quantum circuits (VQCs): Quantum circuits with adjustable parameters are optimised using classical algorithms to perform specific tasks, similar to training neural networks.

Parameterized quantum circuits are trained via classical optimisers, and there is iterative optimisation of circuit parameters.

Quantum neural networks (QNNs): These combine quantum circuits with neural network architectures to create models that potentially outperform classical neural networks on certain tasks.

Quantum gates and circuits mimic neural network layers, and classical optimisation techniques train quantum parameters.

Quantum-classical hybrid algorithms: These are algorithms where quantum and classical computations are interleaved, allowing each to handle parts of the problem they are best suited for.

Here, quantum circuits solve subproblems or enhance classical computations, and classical algorithms manage overall problem structure and optimisation. END

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